A Re-Implementation of LoRA: Low-Rank Adaptation of Large Language Models

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Background & Motivation

The Problem With Full Fine-Tuning

Full fine-tuning requires updating and storing every parameter for each downstream task. At scale this is expensive to train and store.

Goal: Reproduce a subset of Table 2 of Hu et al. (2021) — verify that LoRA matches full fine-tuning accuracy on GLUE using only 0.24% as many trainable parameters.

- **LoRA's hypothesis:** Weight updates ΔW have low intrinsic rank — a small subspace captures the task-relevant adaptation.
- **Core idea**: Freeze W_0 and inject $\Delta W = BA$ where $A \in \mathbb{R}^{(r \times d)}$, $B \in \mathbb{R}^{(d \times r)}$, rank $r << d$.

$$h = W_0x + (\alpha/r) \cdot BAx$$

- **B** = 0 at init → $\Delta W = 0$, model starts identical to pretrained
- **A** random initialization → breaks symmetry so gradients flow
- Applied to W_q and W_v attention matrices → best accuracy/parameter tradeoff

Methodology

Architecture & Datasets

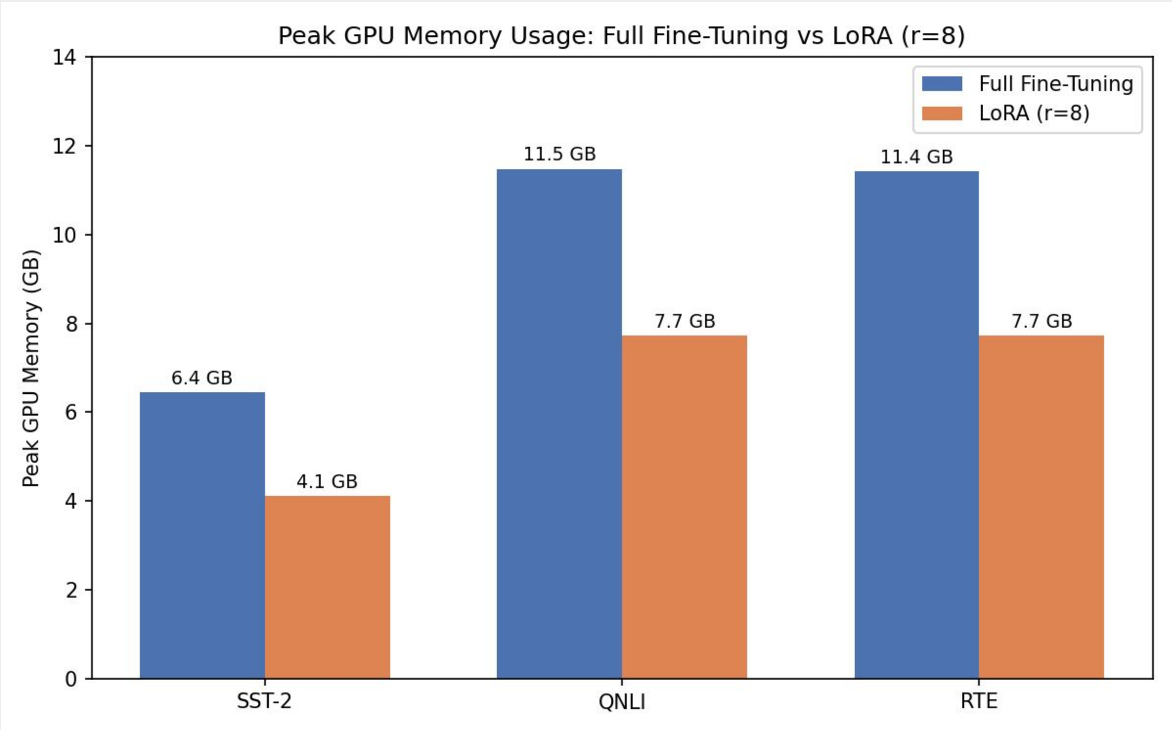
- Load **RobertaForSequenceClassification** (124.6M params) from HuggingFace
- Replace Q, V attention matrices with a custom **LoRALinear** layer module
- Freeze W_0 via requires_grad=False; A, B, classifier head trainable
- **GLUE** tasks: SST-2 (67k), QNLI (105k), RTE (2.5k) · max seq len 512

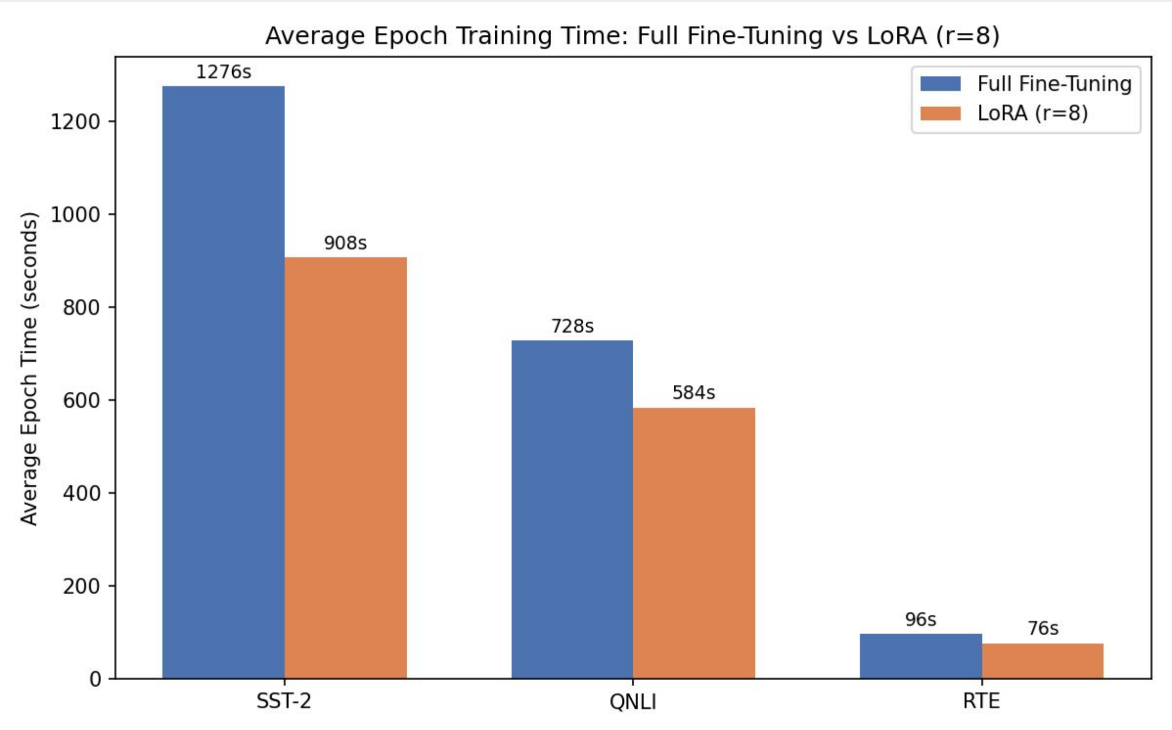
Training & Modifications

- AdamW, weight decay 0.01 (biases/LN exempt), 6% warmup, grad clip 1.0
- LoRA LR: 5e-4 (SST-2/RTE), 4e-4 (QNLI) · Full-FT LR: 2e-5
- **Modification**: Kaiming-uniform initialization for A; N(0,1) caused exploding gradients during training
- **Modification**: Fewer epochs (SST-2: 10 vs 60; QNLI: 10 vs 25); RTE: 80 matches
- **Modification**: RTE trained from scratch; paper started from best MNLI checkpoint

Results

Task	Full FT (ours)	Full FT (paper)	LoRA r=8 (ours)	LoRA (paper)
SST-2	93.69	94.8	**94.38**	95.1
QNLI	92.90	92.8	**92.81**	93.3
RTE	79.42	78.7	**80.14**	86.6





Results - Analysis

- **SST-2 & QNLI**: LoRA within 0.5–0.7 pts of paper despite 6× fewer epochs
- **RTE gap**: 80.14 vs 86.6 can be explained by missing MNLI warm-start
- **33.80%** mean lower peak memory uage across all 3 tasks, **23.11%** mean faster per epoch across all 3 tasks, **99.76%** fewer parameters

Conclusion

- Our re-implementation confirms LoRA's core claim: low-rank adapters matches full fine-tuning accuracy for a fraction of the cost
- Initialization is load-bearing — Kaiming-uniform for A unblocked training entirely.
- LoRA generalizes across task types where different task structures didn't require changing rank or architecture, only the learning rate.

References

- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, L., Wang, W., and Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.
- https://huggingface.co/datasets/nyu-mll/glue
- https://docs.pytorch.org/docs/stable/index.html